# Patrick G. Bissett, Ph. D

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## Employment

Senior Research Scientist, Department of Psychology (2023 –)
Stanford University, Stanford, CA

Research Scientist, Department of Psychology (2019 – 2023)
Stanford University, Stanford, CA

## Education

Postdoctoral Fellow, Department of Psychology (2014 – 2019)
Stanford University, Stanford, CA

Ph. D in Cognition and Cognitive Neuroscience (2009 – 2014)
Vanderbilt University, Nashville, TN

Bachelor of Science with Distinction and Highest Honors in Brain, Behavior, and Cognitive Science (2004 – 2008)
University of Michigan, Ann Arbor, MI

Study abroad program at St. Edmund Hall (2007)
University of Oxford, Oxford, UK

## Publications

Bissett, P. G., Achyutuni, K. G., Li, J. K., Jones, H. M., Shim, S., Rios, J. A. H., Bennett, L. J., Mitchell, M., Jackson, J., Bar Or, M., Roll, D., Meidan, N., Saliy-Grigoryan, A., LePage Drummond, T. M., Pokala, S. S., Cohen, J. R., Schonberg, T., & Poldrack, R. A. (2025). Arbitrary stimuli are not devalued by stopping action: A registered replication of Wessel et al. (2014). *Journal of Experimental Psychology: General, 154,* 2935-2943.

Bissett, P. G., & Poldrack, R. A. (2025). Training with Gordon: Lessons learned in two intersecting scientific journeys. *American Journal of Psychology, 138,* 105-111.

Jaffe, P, I., Santiago-Reyes, G, X., Schafer, R. J., Bissett, P. G., & Poldrack, R. A. (2025). An image-computable model of speeded decision-making. *eLife, 13,* 98351.

Bissett, P. G., Eisenberg, I. W., Shim, S., Rios, J. A. H., Jones, H. M., Hagen, M. P., Enkavi, A, Z., Li, J. K., Mumford, J. A., MacKinnon, D. P., Marsch, L. A., & Poldrack, R. A. (2024). Cognitive tasks, anatomical MRI, and functional MRI data evaluating the construct of self-regulation. *Scientific Data, 11,* 809.

Mumford, J. A., Bissett, P. G., Jones, H. M., Shim, S., Rios, J. A. H., & Poldrack, R. A. (2024). The response time paradox in functional magnetic resonance imaging analyses. *Nature Human Behaviour, 8,* 349-360.

Jaffe, P. I., Poldrack, R. A., Schafer, R. J., & Bissett, P. G. (2023). Modeling human behavior in cognitive tasks with latent dynamic systems. *Nature Human Behaviour, 7,* 986-1000.

Bissett, P. G., Jones, H. M., Hagen, M. P., Bui, T. T., Li, J. K., Rios, J. A. H., Mumford, J. A., Shine, J. M., & Poldrack, R. A. (2023). A dual-task approach to inform the taxonomy of inhibition-related processes. *Journal of Experimental Psychology: Human Perception and Performance, 49,* 277-289.

Bissett, P. G., & Poldrack, R. A. (2022). Estimating the time to do nothing: Toward next-generation models of response inhibition. *Current Directions in Psychological Science, 31,* 556-563.

Walters, J., King, M., Bissett, P. G., Ivry, R. B., Diedrichsen, J., & Poldrack R. A. (2022). Predicting brain activation maps for arbitrary tasks with cognitive encoding models. *NeuroImage, 263,* 119610.

Vaghi, M. M., Hagen, M. P., Jones, H. M., Mumford, J. A., Bissett, P. G., & Poldrack, R. A. (2022). Relating psychiatric symptoms and self-regulation during the COVID-19 crisis. *Translational Psychiatry, 12,* 271.

Scherer, E. A., Metcalf, S. A., Whicker, C. L., Bartels, S. M., Grabinski, M., Kim, S. J., Sweeney, M. A., Lemley, S. M., Lavoie, H., Xie, H., Bissett, P. G., Dallery, J., Michael, K., Lowe, M. R., Onken, L., Proschaska, J. J., Stoeckel, L. E., Poldrack, R. A., MacKinnon, D. P., & Marsch, L. A. (2022). Momentary influences on self-regulation in two populations with health risk behaviors: Adults who smoke and adults who are overweight and have binge-eating disorder. *Frontiers in Digital Health, 4,* 798895.

Bissett, P. G., Jones, H. M., Poldrack, R. A., & Logan, G. D. (2021). Severe violations of independence in response inhibition tasks. *Science Advances, 7,* eabf4355.

Bissett, P. G., Hagen, M. P., Jones, H. M., & Poldrack, R. A. (2021). Design issues and solutions for stop-signal data from the Adolescent Brain Cognitive Development [ABCD] study. *eLife, 10,* e60185.

Zmigrod, L., Eisenberg, I. W., Bissett, P. G., Robbins, T. W., & Poldrack, R. A. (2021). The cognitive and perceptual correlates of ideological attitudes: A data-driven approach. *Philosophical Transactions of the Royal Society B, 376,* 20200424.

Mazza, G. L., Smyth, H. L., Bissett, P. G., Canning, J. R., Eisenberg, I. W., Enkavi, A. Z.,… & MacKinnon, D. P. (2021). Correlation database of 60 cross-disciplinary surveys and cognitive tasks assessing self-regulation. *Journal of Personality Assessment, 103,* 238-245.

Thompson, W. H., Wright, J., Bissett, P. G., & Poldrack, R. A. (2020). Dataset decay and the problem of sequential analyses on open dataset. *eLife, 9*, e53498.

Thompson, W. H., Wright, J., & Bissett, P. G. (2020). Open Exploration. *eLife, 9,* e52157.

Enkavi, A. Z., Eisenberg, I. W., Bissett, P. G., Mazza, G. L., MacKinnon, D. P., Marsch, L. A., & Poldrack, R. A. (2019). Reply to Friedman and Banich: Right measures for the research question. *Proceedings of the National Academy of Sciences, 116,* 24398-24399.

Eisenberg, I. W., Bissett, P. G., Enkavi, A. Z., Li, J., MacKinnon, D. P., Marsch, L. A., & Poldrack, R. A. (2019). Uncovering the structure of self-regulation through data-driven ontology discovery. *Nature Communications, 10,* 2319.

Verbruggen, F., Aron, A. R., Band, G. P. H., Beste, C., Bissett, P. G., Brockett, A. T.,… & Boehler, C. N. (2019). A consensus guide to capturing the ability to inhibit actions and impulsive behaviors in the stop-signal task. *eLife, 8,* e46323.

Enkavi, A. Z., Eisenberg, I. W., Bissett, P. G., Mazza, G. L., MacKinnon, D. P., Marsch, L. A., & Poldrack, R. A. (2019). A large-scale analysis of test-retest reliabilities of self-regulation measures. *Proceedings of the National Academy of Sciences, 116,* 5472-5477.

Eisenberg, I. W., Bissett, P. G., Canning, J. R., Dallery, J., Enkavi, A. Z., Whitfield-Gabrieli, S.,… & Poldrack, R. A. (2018). Applying novel technologies and methods to inform the ontology of self-regulation. *Behaviour Research and Therapy, 101,* 46-57.

Wylie, S. A., van Wouwe, N. C., Godfrey, S. G., Bissett, P. G., Logan, G. D., Kanoff, K. E.,… & van den Wildenberg, W. P. M. (2018). Dopaminergic medication shifts the balance between going and stopping in Parkinson's disease. *Neuropsychologia, 109,* 262-269.

Bissett, P. G., Grant, L. D., & Weissman, D. H. (2017). Resisting distraction and response inhibition trigger similar enhancements in future performance. *Acta Psychologica, 180,* 40-51.

Weissman, D. H., Colter, K. M., Grant, L. D., & Bissett, P. G. (2017). Identifying stimuli that cue multiple responses triggers the congruency sequence effect independent of response conflict. *Journal of Experimental Psychology: Human Perception and Performance, 43,* 677-689.

Shine, J. M., Bissett, P. G., Bell, P. T., Koyejo, O., Balsters, J. H., Gorgolewski, K. J., Moodie, C. A., & Poldrack, R. A. (2016). The dynamics of functional brain networks: Integrated network states during cognitive task performance. *Neuron, 92,* 544-554.

Georgiades, M. J., Gilat, M., Ehgoetz Martens, K. A.,Walton, C. C., Bissett, P. G., Shine, J. M., & Lewis, S. J. (2016). Investigating motor initiation and inhibition deficits in patients with Parkinson's disease and freezing of gait using a virtual reality paradigm. *Neuroscience, 337,* 153-162.

Sochat, V. V., Eisenberg, I. W., Enkavi, A. Z., Li, J., Bissett, P. G., & Poldrack. R. A. (2016). The experiment factory: Standardizing behavioral experiments. *Frontiers in Psychology, 7,* 1-9.

Bissett, P. G., Logan, G. D., van Wouwe, N. C., Tolleson, C. M., Phibbs, F . T., Claassen, D. O., & Wylie, S. A. (2015). Generalized motor inhibitory deficit in Parkinson's disease patients who freeze. *Journal of Neural Transmission, 122,* 1693-1701.

Bissett, P. G., & Logan, G. D. (2014). Selective Stopping? Maybe not. *Journal of Experimental Psychology: General, 143,* 455-472.

Bissett, P. G., & Logan, G. D. (2013). Stop before you leap: Changing eye and hand movements requires stopping. *Journal of Experimental Psychology: Human Perception and Performance, 39,* 941-946.

Bissett, P. G. (2013). The countermanding task revisited: Mimicry of race models. [Commentary on "The countermanding task revisited: Fast stimulus detection is a key determinant of psychophysical performance"]. *The Journal of Neuroscience, 33,* 12150-12151.

Bissett, P. G., & Logan, G. D. (2012). Post-stop-signal adjustments: Inhibition improves subsequent inhibition. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 38,* 955-966.

Bissett, P. G., & Logan, G. D. (2012). Post-stop-signal slowing: Strategies dominate reflexes and implicit learning. *Journal of Experimental Psychology: Human Perception and Performance, 38,* 746-757.

Yamaguchi, M., Logan, G. D., & Bissett, P. G. (2012). Stopping while going! Response inhibition does not suffer dual-task interference. *Journal of Experimental Psychology: Human Perception and Performance, 38,* 123-134.

Bissett, P. G., & Logan, G. D. (2011). Balancing cognitive demands: Control adjustments in the stop-signal paradigm. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 37,* 392-404.

Bissett, P. G., Nee, D. E., & Jonides, J. (2009). Dissociating Interference-Control Processes between Memory and Response. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 35,* 1306-1316.

Cooper, S., Iyer, G., Tarquini, M., & Bissett, P., (2006). Nocodazole does not synchronize cells: implications for cell-cycle control and whole-culture synchronization. *Cell and Tissue Research, 324,* 237-242.

## Preprints

Bissett, P. G., Achyutuni, K. G., Rios, J. A. H., Jones, H. M., Bennett, L, J., & Poldrack, R. A. (submitted). Sharing control with a machine impairs human response inhibition. https://osf.io/preprints/psyarxiv/vkrqd_v2

Hagen, M. P., Provins, C., MacNicol, E., Li, J. K., Gomez, T., Garcia, M., Seeley, S. H., Legarreta, J. H., Norgaard, M., Bissett, P. G., Poldrack, R. A., Rokem, A., & Esteban, O. (submitted). Quality assessment and control of unprocessed anatomical, functional, and diffusion MRI of the human brain using MRIQC. https://www.biorxiv.org/content/10.1101/2024.10.21.619532v1

Vaghi, M. M., Norgaard, M., Shim, S., Rios, J. A. H., Bissett, P. G., Rodriguez, C. I., Poldrack. R. A. (submitted). Frontostriatal salience network expands as executive networks contract in Obsessive-Compulsive Disorder. https://www.biorxiv.org/content/10.1101/2025.05.21.654808v1

## Selected Conference Presentations

Bissett, P. G., Achyutuni, K. G., Rios, J. A. H., Jones, H. M., Bennett, L. J., & Poldrack. R. A. Shared control impairs cognitive control: Human response inhibition slows when machines fail to inhibit. Annual Meeting of the Cognitive Science Society. San Francisco, California. Amsterdam, July 30-August 2, 2025.

Bissett, P. G., Shim, S., Bennett, L. J., Rios, J. A. H., Jones, H. M., Hagen, M. P., Achyutuni, K. G., Demidenko, M. I., Mumford, J. A., Shine, J. M., & Poldrack. R. A. Characterizing cognitive control networks using a precision neuroscience approach. Annual Meeting of the Organization of Human Brain Mapping. Brisbane, Australia. June 24-28, 2025.

Bissett, P. G., Shim, S., Rios, J. A. H., Mumford, J. A., & Poldrack. R. A. Cue switching but not task switching drives fronto-parietal activity in cued task switching paradigm. Annual Meeting of the Organization of Human Brain Mapping. Seoul, Korea. June 23-27, 2024.

Bissett, P. G., Bennett, L. J., Rios, J. A. H., Shim, S., McKee, P. C., Iyer, C. S., Ram, N., & Poldrack. R. A. Mapping task measures to latent constructs: An expert survey of the NIMH RDoC cognitive domain. Annual Meeting of the Organization of Human Brain Mapping. Seoul, Korea. June 23-27, 2024.

Bissett, P. G., Shim, S., Rios, J. A. H., Jones, H. M., Hagen, M. P., Mumford, J. A., Shine, J. M., & Poldrack. R. A. Dual-task costs relate to overlap in single-task fMRI maps: A dense scanning study. Annual Meeting of the Organization of Human Brain Mapping. Montreal, Canada. June 22-26, 2023.

Bissett, P. G., Shim, S., Rios, J. A. H., Jones, H. M., Hagen, M. P., Mumford, J. A., Shine, J. M., & Poldrack. R. A. Characterizing cognitive control networks using a precision neuroscience approach. Annual Meeting of the Cognitive Neuroscience Society. San Francisco, California. March 25-28, 2023.

Bissett, P. G., Jones, H. M., Hagen, M. P., Bui, T. T., Li, J. K., Rios, J. A. H., Mumford, J. A., Shine, J. M., & Poldrack, R. A. A dual-task approach to inform the taxonomy of inhibition-related processes. Annual Meeting of the Organization of Human Brain Mapping. Glasgow, United Kingdom. June 19-23, 2022.

Bissett, P. G., Eisenberg, I. W., Jones, H. M., Hagen, M. P., Rios, J. A. H., Shim, S., Enkavi, A. Z., Li, J. K., Mumford, J. A., Shine, J. M., MacKinnon, D. P., Marsch, L. A., & Poldrack, R. A. Toward a neuro-cognitive ontology of self-regulation. Annual Meeting of the Organization of Human Brain Mapping. Glasgow, United Kingdom. June 19-23, 2022.

Bissett, P. G., Eisenberg, I. W., Jones, H. M., Hagen, M. P., Rios, J. A. H., Shim, S., Enkavi, A. Z., Li, J. K., Mumford, J. A., Shine, J. M., MacKinnon, D. P., Marsch, L. A., & Poldrack, R. A. Toward a neuro-cognitive ontology of self-regulation. Annual Meeting of the Cognitive Neuroscience Society. San Francisco, California. April 23-26, 2022.

Bissett, P. G., Hagen, M. P., Jones, H. M., & Poldrack, R. A. Design issues and solutions for Stopping Data from the Adolescent Brain Cognitive Development (ABCD) study. Annual Meeting of the Organization of Human Brain Mapping. Virtual Event, June 21-25, 2021.

Bissett, P. G., Poldrack, R. A., & Logan, G. D. Systematic violations of independence in models of response inhibition. Annual Meeting of the Organization of Human Brain Mapping. Rome, Italy, June 9-13, 2019.

Bissett, P. G., Poldrack, R. A., & Logan, G. D. Systematic violations of independence in models of response inhibition. Association for Psychological Science Annual Convention. Washington DC, May 23-26, 2019.

Bissett, P. G., Poldrack, R. A., & Logan, G. D. Severe violations of independence in response inhibition tasks are pervasive and consequential. Annual Meeting of Control Processes. Providence, Rhode Island, May 16-18, 2019.

Bissett, P. G., Shine, J. M., Durnez, J., Li, J., & Poldrack. R. A. Combining stopping and working memory slows response inhibition and relates to network integration. Annual Meeting of the Organization of Human Brain Mapping. Singapore, June 17-21, 2018.

Bissett, P. G., Eisenberg, I. W., & Poldrack, R. A. Reactive, proactive, and selective stopping: What can 500,000 trials tell us about inhibitory control? Annual Meeting of the Psychonomic Society. Vancouver, Canada, November 9-12, 2017.

Bissett, P. G., Shine, J. M., Durnez, J., Li, J., Gorgolewski, K. J., Esteban, O., Blair, R., & Poldrack. R. A. Behavioral and neural interactions between working memory and stop-signal inhibition. Annual Meeting of the Organization of Human Brain Mapping. Vancouver, Canada, June 25-29, 2017.

Bissett, P. G., Shine, J. M., Koyejo, O., Gorgolewski, K. J., & Poldrack, R. A. Fluctuation in network topology predict N-back correct and errors as a function of cognitive load. Annual Meeting of the Organization of Human Brain Mapping. Geneva, Switzerland, June 26-30, 2016.

Bissett, P. G., & Poldrack, R. A. Fast inhibition or impulsive inhibition? Towards a common threshold for engaging and inhibiting responses. Association for Psychological Science Annual Convention. Chicago, Illinois, May 26-29, 2016.

Bissett, P. G., Logan, G. D., & Wylie, S. A. Selective inhibition in young adults, older adults, and Parkinson's Disease. Annual Meeting of the Society for Neuroscience. San Diego, California, November 9-13, 2013.

Bissett, P. G., & Logan, G. D. Strategies for selective stopping: Challenging the race model. Annual Meeting of the Psychonomic Society. Minneapolis, Minnesota, November 15-18, 2012.

Bissett, P. G., & Logan, G. D. Post-stop-signal slowing: Strategies dominate reflexes and implicit learning. Annual Meeting of the Psychonomic Society. Seattle, Washington, November 3-6, 2011.

Bissett, P. G., Nee, D. E., & Jonides, J. Interference mechanisms for response production and memory representations. Annual Meeting of the Psychonomic Society. Houston, Texas, November 16-19, 2006.

## Invited Talks

Measuring latent constructs: Lessons from ontology development for human cognition. Symposium on "What are we measuring in AI Systems?". Stanford University, California, November 14, 2025.

Using expert knowledge and data-driven analyses to improve the ontology of cognition. Annual Meeting of the Cognitive Science Society. Rotterdam, Amsterdam, July 23-27, 2024.

Toward a new model for response inhibition and updating the taxonomy of cognitive control. University of Nevada, Reno Cognitive and Brain Sciences early career seminar. April 18, 2024.

What is the structure of "self-regulation"? Association for Psychological Science Annual Convention. Washington DC, May 23-26, 2019.

Applying novel technologies and methods to inform the ontology of self-regulation. The NIH Science of Behavior Change Research Network Annual Meeting. Bethesda, Maryland, January 10, 2019.

Proactive control improves reactive response inhibition. Annual Meeting of Control Processes. Amsterdam, The Netherlands, October 11-13, 2017.

Strategies for selective stopping. Translational Aspects of Stopping Workshop. San Diego, California, November 8, 2013.

Selective response inhibition: Basic, clinical, developmental, and neuroscientific implications. University of Texas at Austin, Texas, October 7, 2013.

Selective stopping strategies across effectors, age, and Parkinson's Disease. University of California at San Diego, California, July 24, 2013.

## Awards

APA Editor's Choice Award at *Journal of Experimental Psychology: General* given to our paper titled: Arbitrary stimuli are not devalued by stopping action: A registered replication of Wessel et al. (2014).

Association for Psychological Science Rising Star Award, 2021

Vanderbilt University Graduate Student Travel Award, 2011, 2012, 2013

National Science Foundation Graduate Research Fellowship Honorable Mention, 2009

Vanderbilt University Departmental Award in recognition of exemplary academic record, 2009

University of Michigan W. B. Pillsbury Prize for Most Outstanding Honors Thesis, 2008

University of Michigan degree conferred with "Highest Honors" given for the best senior honors thesis in the Psychology Department, 2008

University of Michigan Spike Tanner Memorial Award for Research, 2007

University of Michigan Eli Lilly Research and Travel Award, 2006

National Merit Scholarship Finalist, 2003

## Current Funding

National Institute of Mental Health (R01), 8/19/2022-5/31/2027:
Data-driven validation of cognitive RDoC dimensions using deep phenotyping
PI: Russell A. Poldrack, Role: Co-Investigator/Co-Author
$3,801,968

## Previous Funding

National Institutes of Health (U24) 9/15/2020-9/14/2025:
Columbia University Science of Behavior Change Resource and Coordinating Center renewal
PI: Donald Edmondson, Role: Co-Investigator
$6,239,458

National Institute of Mental Health (R01), 7/1/2018-3/31/2024:
Characterizing cognitive control networks using a precision neuroscience approach.
PI: Russell A. Poldrack, Role: Co-Investigator/Co-Author
$2,097,193

Ruth L. Kirschstein National Research Service Award from the National Institute of Drug Abuse (F32), 4/1/2016-3/31/2019:
Elucidate the mechanisms underlying inhibition induced devaluation.
PI: Patrick G. Bissett
$179,218

Stanford Center for Cognitive and Neurobiological Imaging (CNI) Innovations Grant, 2/15/18-8/31/18:
Characterizing cognitive control networks using a precision neuroscience approach
PI: Patrick G. Bissett
$2,000

Stanford Center for Cognitive and Neurobiological Imaging (CNI) Innovations Grant, 2/1/17-8/15/17:
Behavioral and neural interactions between working memory and stop-signal inhibition
PI: Patrick G. Bissett
$2,835

## Peer Review

*Ad hoc journal reviewer:* Acta Psychologica, American Journal of Psychology, Behavior Research Methods, Behavioural Brain Research, Cerebral Cortex, Child Development, Cognition, Cognitive Development, Cognitive Psychology, Communications Psychology, Cortex, Developmental Science, eLife, eNeuro, European Journal of Neuroscience, Experimental Brain Research, Human Brain Mapping, Imaging Neuroscience, JAMA Psychiatry, Japanese Psychological Research, Journal of Applied Research in Memory and Cognition, Journal of Cognitive Neuroscience, Journal of Experimental Child Psychology, Journal of Experimental Psychology: General, Journal of Experimental Psychology: Human Perception and Performance, Journal of Neurophysiology, Memory & Cognition, Nature Human Behaviour, Neurobiology of Aging, Neuroscience and Biobehavioral Reviews, NeuroImage, Neuropsychologia, Personality and Individual Differences, PLOS ONE,

Psychological Research, Psychological Science, Psychonomic Bulletin & Review, Quarterly Journal of Experimental Psychology, Scientific Data, Scientific Reports, The Journal of Neuroscience, Trends in Cognitive Sciences, Vision Research

*Ad hoc grant reviewer:* Medical Research Council

*Ad hoc conference abstract reviewer:* Annual Meetings of the Organization for Human Brain Mapping, Conference on Cognitive Computational Neuroscience

## Mentoring

Supervisor or co-supervisor to over 40 research coordinators, research assistants, graduate students, and postdocs.

## Teaching Experience

-Guest lecturer in Psych 252 (Statistical Methods) at Stanford University, March 2022

-Graduate Teaching Assistant in Experimental Design (x2), Introduction to Psychology, and Perception at Vanderbilt University

## Professional Societies

Association for Psychological Science, Cognitive Science Society, Organization for Human Brain Mapping, Psychonomic Society, Society for Neuroscience

## Media Appearances

Bissett, P. G. (May 25, 2025). Invited Guest “What happens when AI takes the wheel?” on *RemAIning Human* (Video Podcast**).** [https://ceciliacallas.substack.com/p/what-happens-when-ai-takes-the-wheel]